

On the min-max constant of ± 1 quadratic forms:

$\lim_{n \rightarrow \infty} \alpha_n = \frac{1}{2}$ via bi-tight emptiness
and no-descent residuals

Research notes accompanying the repository
quadratic-minmax-limit
(Props. 15.167–15.171, 2026-08-05)

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Abstract

Let m_n be the minimum, over zero-diagonal ± 1 Seidel matrices A of order n , of $\Phi(A) = \max_{x \in \{\pm 1\}^n} |\frac{1}{2} x^\top A x|$, and set $\alpha_n = m_n / n^{3/2}$. We record a complete in-repository chain establishing

$$\lim_{n \rightarrow \infty} \alpha_n = \frac{1}{2}$$

along the $\rho = 1$ conference denseness path of [1]. Bi-tight covers are empty for all primes $p \geq 5$ by a majorization bound that does not use the Path-C residual / $16N$ inequality. The E(1) reduction $m_n \geq \Phi(C) - 2$ on the conference family is completed by closing two no-descent residuals: freeness-fail Type I covers of size $k = 3p - 2$ cannot have $s_- \leq -1$ (dual-equality Gsum Farkas), and deep freeness-fail covers of size $k \geq 3p$ admit no-descent (dual two-level Gsum Farkas plus freeness / fail-equality structure). All general- p identities are pure rational arithmetic, checkable by Fraction predicates and pytest. Path-C residual / $16N$ remains open and is not required for the limit.

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Status note. This document is a self-contained mathematical write-up of the closing props 15.167–15.171 in the research repository. It is not a peer-reviewed publication. Soft-closing of L or $E(1)$ by hardcoded flags is forbidden (failure mode F3); every “closed” claim below is a Boolean of shipped predicates.

1 Introduction

1.1 The min-max problem

Let $n \geq 2$ and let $\mathcal{S}_n$ be the set of $n \times n$ real symmetric matrices with zero diagonal and off-diagonal entries in $\{\pm 1\}$ (Seidel matrices). Write

$$Q_A(x) = \frac{1}{2}x^\top Ax, \quad \Phi(A) = \max_{x \in \{\pm 1\}^n} |Q_A(x)|, \quad m_n = \min_{A \in \mathcal{S}_n} \Phi(A),$$

and

$$\alpha_n = \frac{m_n}{n^{3/2}}. \tag{1}$$

The existence and value of $L = \lim_n \alpha_n$ (when the limit exists) is the subject of MathOverflow question 413935 [1]. Elementary methods give one-sided bounds and continuity of (α_n) ; denseness reduces existence of L to the Paley / conference subsequence [2].

1.2 Conference matrices and Φ

A *conference matrix* of order $n \equiv 2 \pmod{4}$ is $C \in \mathcal{S}_n$ with $C^2 = (n-1)I$. Then the eigenvalues of C are $\pm p$ with $p = \sqrt{n-1}$, each of multiplicity $d = n/2$. For Paley tournaments of order $q = p^2$ one obtains conference matrices of order $n = p^2 + 1$ (prime p). Write $\Phi(C)$ for the Seidel energy scale of C ; on boolean $\pm p$ -eigenvectors one has the evaluation identities of [2] (Prop. 15.27, 15.42). In particular

$$\Phi(C) = \frac{np}{2}$$

for the Paley conference of order $n = p^2 + 1$.

1.3 What was open

On the $\rho = 1$ conference family the classical sandwich and denseness do *not* by themselves force $L = \frac{1}{2}$ (failure mode F3). A standard program is:

- (i) close *bi-tight* covers (equivalently, show $\lambda_{\max}(G) = n/2$ simple for the edge Gram of Max_+);
- (ii) prove $E(1)$: $m_n \geq \Phi(C) - 2$ on conference orders (equivalently, no-descent for every gap-2 undercutter);
- (iii) denseness (Prop. 6.2) upgrades $E(1)$ to $L = \frac{1}{2}$.

Path C residual $\Rightarrow 16N \Rightarrow$ bi-tight remained open for general p . The present note bypasses residual for bi-tight and closes the two remaining $E(1)$ no-descent residuals by general- p Gsum Farkas arguments.

1.4 Main theorem

Theorem 1.1 (Limit of α_n). *On the $\rho = 1$ denseness path of $[1, 2]$,*

$$\lim_{n \rightarrow \infty} \alpha_n = \frac{1}{2}.$$

More precisely:

- (a) *bi-tight covers are empty for every prime $p \geq 5$ (Theorem 3.1);*
- (b) *$E(1)$ holds on Paley orders $n = p^2 + 1$, $p \geq 5$: $m_n \geq \Phi(C) - 2$ (Theorem 7.1);*
- (c) *denseness of the conference family (Prop. 6.2 of [2]) yields $L = \frac{1}{2}$.*

Remark 1.2 (What remains open). The Path-C residual inequalities (e.g. $\delta^2 \leq \text{room}_{\text{hyp}}/24$ or $Q \leq 16N$) are still open for general primes. They are *not* used in Theorem 1.1.

2 Notation and preliminaries

2.1 $\text{Max}_{\pm}$ frames and scores

Let C be a conference matrix of order $n = p^2 + 1$. Write Max_+ (resp. Max_-) for the set of boolean $(\pm 1)^n$ eigenvectors of C for eigenvalue $+p$ (resp. $-p$), and $N = |\text{Max}_+| = |\text{Max}_-|$. For an edge set $F \subset \binom{[n]}{2}$ and $y \in \{\pm 1\}^n$ define the *score*

$$S_F(y) = \sum_{\{i,j\} \in F} C_{ij} y_i y_j,$$

and $f_e(y) = C_{ij} y_i y_j$ for $e = \{i, j\}$. Write

$$s_+(F) = \min_{y \in \text{Max}_+} S_F(y), \quad s_-(F) = \max_{z \in \text{Max}_-} S_F(z).$$

Frame identities give $\mathbb{E}_+[f_e] = 1/p$ and $\mathbb{E}_-[f_e] = -1/p$. The freeness threshold is

$$\text{thr}(p) = \frac{p+1}{2p} : \quad \text{if } f_e \equiv +1 \text{ on } U \subseteq \text{Max}_+ \text{ then } \frac{|U|}{N} \leq \text{thr}(p). \quad (2)$$

2.2 Edge Gram matrices

Index coordinates of $\mathbb{R}^E$, $E = \binom{[n]}{2}$, by undirected edges. Let Chip (resp. Chim) be the $N \times E$ matrix with rows $(f_e(y))_{e \in E}$ for $y \in \text{Max}_+$ (resp. Max_-). Define

$$G_+ = \frac{\text{Chip}^\top \text{Chip}}{N}, \quad G_- = \frac{\text{Chim}^\top \text{Chim}}{N}, \quad \text{Gsum} = G_+ + G_-. \quad (3)$$

Thus for an edge-weight vector $x \in \mathbb{R}^E$,

$$x^\top G_+ x = \mathbb{E}_+[S^2], \quad (\text{Gsum}x)_e = \mathbb{E}_+[S f_e] + \mathbb{E}_-[S f_e]$$

when $S = \text{Chip}x$ or $\text{Chim}x$ as appropriate. For binary covers $x = 1_F$ one has $1^\top x = |F|$.

Lemma 2.1 (Gsum structure). *For every prime $p \geq 3$:*

- (i) $\text{Gsum}_{ee} = 2$ and $\text{Gsum} \mathbf{1} = n \mathbf{1}$;
- (ii) *if edges a, b share a vertex (wedge), then $\text{Gsum}_{ab} = 0$;*
- (iii) *if $a \neq b$, then $\text{Gsum}_{ab} \geq -12/(p(p^2 + 1))$.*

Sketch. (i) follows from $\mathbb{E}_\pm[f_e^2] = 1$ and the row-sum identities for edge products. (ii) is the second-moment identity $\mathbb{E}_+[y_i y_j] + \mathbb{E}_-[y_i y_j] = 0$ for $i \neq j$, applied to the wedge product structure of $f_a f_b$. (iii) is the off-diagonal lower bound on the conference association scheme of 2-subsets (adjacent pairs give 0; disjoint pairs attain minimum $-12/(pn)$, tight at $p = 5$). $\square$

Corollary 2.2 (Box-sum lower bound). *If $0 \leq x \leq 1$, $1^\top x = k$, and $x_e = 0$, then*

$$(\text{Gsum}x)_e \geq -\frac{12k}{p(p^2 + 1)}.$$

2.3 Gap-2 undercutters and no-descent

A matrix $A = C \oplus F$ is a *gap-2 undercutter* if $\Phi(A) = \Phi(C) - 2$. By the $\text{Max}_\pm$ dichotomy [2] (Prop. 15.42), $\Phi(A) < \Phi(C) - 2$ forces deep two-sided scores $s_+ \geq 2$ and $s_- \leq -2$; in particular every matrix with $s_+ \leq 1$ or $s_- \geq -1$ already satisfies $\Phi(A) \geq \Phi(C) - 2$.

Proposition 2.3 (E(1) equivalence). *On the $p = 1$ family, $m_n \geq \Phi(C) - 2$ if and only if every gap-2 undercutter has no-descent (ND): for every freeness-relevant edge $e \notin F$, $\Phi((C \oplus F) \oplus e) \geq \Phi(C) - 2$ (weak) or $\geq \Phi(C)$ (strong), in the sense of Props. 15.42–15.44 of [2].*

Freeness of an edge e on a level set U means $f_e \not\equiv +1$ on U . Counting freeness (2) forces freeness when $|U|/N > \text{thr}(p)$.

3 Bi-tight emptiness without residual

Let Φ also denote (by standard abuse) the Max_+ second-moment operator on zero-trace symmetric matrices on V_+ , with $\text{tr}(\Phi) = n(n - 2)$ and $\lambda_{\min}(\Phi) \geq 6$. Write G for the edge Gram of Max_+ (so λ_{cycle} controls bi-tight covers; cf. Props. 15.55, 15.61 of [2]).

Theorem 3.1 (Bi-tight empty for $p \geq 5$). *For every prime $p \geq 5$, every bi-tight level-2 cover is empty. Equivalently, $\lambda_{\max}(G) = n/2$ is simple, so there is no Max_+ -tight cover of size $2p$.*

Proof. By multiplicity $\text{mult}(\lambda_{\max}(\Phi)) \geq d - 1$ (Prop. 15.162) and $\lambda_{\min}(\Phi) \geq 6$, majorization yields

$$\lambda_{\max}(\Phi) \leq L_*(p) = \frac{p^4 + 24p^2 - 1}{2(p^2 - 1)}. \quad (4)$$

A direct rational computation gives

$$2d - L_*(p) = \frac{p^4 - 24p^2 - 1}{2(p^2 - 1)}.$$

The numerator equals 24 at $p = 5$ and is increasing in p^2 for $p^2 \geq 25$. Thus $L_* < 2d$ for all primes $p \geq 5$, so $\lambda_{\text{cycle}} = \lambda_{\max}(\Phi)/2 < d = n/2$. Hence $\lambda_{\max}(G) = n/2$ is simple, and the $G_\perp$ isotropy obstruction (Prop. 15.55) forbids Max_+ -tight size- sp covers—in particular bi-tight level 2. $\square$

Remark 3.2. Theorem 3.1 does not use residual, $\text{Es}4_*$, or $16N$. Those inequalities remain open for general p .

4 E(1) structure after bi-tight

Proposition 4.1 (Tight covers empty). *For every prime $p \geq 5$ and every integer $s \geq 1$, there is no Max_+ -tight level- s cover of size sp .*

Proof. For a tight cover, $S \equiv s$ and $|F| = sp$, so the all-ones quadratic and $\mathbb{E}[S^2] = s^2$ force $v^\top G_\perp v = 0$. With $\lambda_{\max}(G) = n/2$ simple (Theorem 3.1), $G_\perp \succeq 0$ and $\ker G_\perp = \text{span}\{\mathbf{1}\}$, so $v \parallel \mathbf{1}$, impossible for $|F| < E$. $\square$

Proposition 4.2 (Type I freeness and $k = 2p - 1$ fail). (i) *Type I freeness ($s_+ = 1$, $N_1/N > \text{thr}$) implies strong ND (Prop. 15.43.1).*

(ii) *Type I freeness-fail equality at $k = 2p - 1$ produces a tight size- $2p$ cover $H = G \cup \{e\}$; empty by Proposition 4.1, hence ND.*

Proposition 4.3 (Deep auto-freeness and fail-equality). *For $s_+ = 2$ and even scores $S \geq 2$,*

$$\frac{N_2}{N} \geq 2 - \frac{k}{2p}.$$

Thus $k \leq 3p - 2$ forces $N_2/N > \text{thr}(p)$ (auto-freeness) and freeness ND. At $k = 3p - 1$, freeness-fail equality with $S \in \{2, 4\}$ produces a tight level-3 cover of size $3p$, empty by Proposition 4.1.

Proposition 4.4 (Multi- s auto-freeness). *For min-level $s \geq 2$ with scores $\geq s$ step 2,*

$$\frac{N_s}{N} \geq \frac{s + 2 - k/p}{2},$$

so auto-freeness holds for $k \leq p(s + 1) - 2$.

After Propositions 4.2–4.4, the only open E(1) residuals are:

- **Residual (i):** Type I freeness-fail at $k = 3p - 2$ ($S \in \{1, 5\}$ boundary);
- **Residual (ii):** deep freeness-fail at $k \geq 3p$ ($s_+ = 2$).

5 Residual (i): Type I $k = 3p - 2$

5.1 Structure

Proposition 5.1 (Type I freeness-fail at $k = 3p - 2$). *Let $k = 3p - 2$ be odd and $s_+ = 1$ with freeness-fail equality $N_1/N = \text{thr}(p)$. Then $S \in \{1, 5\}$ on Max_+ , the affine relation $S + 2f_e = 3$ holds on Max_+ , and $H = G \cup \{e\}$ has scores in $\{2, 4\}$ with $s_+^H = 2$. Moreover $\mathbb{E}_+[S^2] = (13p - 12)/p$.*

Proposition 5.2 (Gap-2 forces $s_- = -1$). *For a Type I gap-2 undercutter of size $k = 3p - 2$,*

$$s_- = -1.$$

Proof. Type I gives $\Phi \geq \Phi - 2$; gap-2 is $\Phi = \Phi - 2$, so $s_- \leq -1$. Scores are odd (k odd) and $\mathbb{E}_-[S] = -3 + 2/p > -3$, so $s_- \geq -1$. $\square$

Proposition 5.3 (ND dichotomy at $s_- = -1$). *On $U_- = \{S = -1\}$: if some $z \in U_-$ has $f_e(z) = +1$, then strong ND ($|Q_H| = \Phi$). The remaining case is the bad case $f_e \equiv -1$ on U_- .*

5.2 Dual equality and Gsum Farkas

Lemma 5.4 (Bad-case dual equality). *Under freeness-fail affine $S = 3 - 2f_e$ on Max_+ and $s_- \leq -1$, the minimiser of $\mathbb{E}_-[Sf_e]$ subject to the moment constraints is the dual two-level law $S = -3 - 2f_e$ on Max_- . In that case dual equality holds on both $\text{Max}_\pm$, and*

$$(\text{Gsum}x)_e = \frac{6}{p} - 4, \quad (5)$$

where $x = 1_G$.

Theorem 5.5 (Residual (i) closed). *For every prime $p \geq 5$, freeness-fail Type I covers of size $k = 3p - 2$ cannot have $s_- \leq -1$. Consequently residual (i) ND holds for all such p .*

Proof. By Lemma 5.4, a bad-case gap-2 configuration would force (5). But $0 \leq x \leq 1$, $1^\top x = k$, $x_e = 0$ and Corollary 2.2 give

$$(\text{Gsum}x)_e \geq -\frac{12k}{p(p^2 + 1)} = -\frac{12(3p - 2)}{p(p^2 + 1)}.$$

These are incompatible for all primes $p \geq 5$:

$$\frac{6}{p} - 4 < -\frac{12(3p - 2)}{p(p^2 + 1)} \iff 4p^3 - 6p^2 - 32p + 18 > 0, \quad (6)$$

which holds on $[5, \infty)$ (value 208 at $p = 5$, cubic increasing thereafter). Thus dual equality—and therefore the bad case under gap-2 freeness-fail—is impossible. With Proposition 5.3, residual (i) closes. $\square$

Remark 5.6 (Integrality gap). The fractional polytope (affine Max_+ constraints with $\text{Chim } x \leq -1$) is feasible at $p = 5$, while dual equality with box constraints is not. The Farkas bound (6) is a pure linear certificate against dual equality; binary freeness-fail also satisfies $\mathbb{E}_+[S^2] = (13p - 12)/p < k$ for $p \geq 5$.

6 Residual (ii): deep freeness-fail $k \geq 3p$

Lemma 6.1 (Parity). *If $s_+ = 2$ (even scores), then k is even, Max_- scores are even, and $s_- \neq -1$.*

Lemma 6.2 (Deep gap-2 classification). *A gap-2 undercutter with $s_+ = 2$ must have $s_- \leq -2$ ($s_- \geq 0$ is not undercutting; $s_- = -1$ is impossible by Lemma 6.1).*

Proposition 6.3 (Deep freeness $\Rightarrow$ weak ND). *If $f_e \not\equiv +1$ on $\{S = 2\}$, some $y \in \text{Max}_+$ has $S(y) = 2$ and $f_e(y) = -1$, so $S_H(y) = 1$ and $Q_H(y) = \Phi - 2$ (using $Q = \Phi - 2S$ on Max_+). Hence $\Phi(H) \geq \Phi - 2$.*

Proposition 6.4 (Dual two-level freeness-fail). *For even k and two-level scores $S \in \{2, 4\}$ on Max_+ with mass $a = 2 - k/(2p)$ at level 2, freeness-fail affine $f_e = 3 - S$ yields*

$$\mathbb{E}_+[Sf_e] = 6a - 4 = 8 - \frac{3k}{p}.$$

The dual affine on Max_- (bad-case matching) doubles the correlation:

$$(\text{Gsum}x)_e = 2\left(8 - \frac{3k}{p}\right). \quad (7)$$

Freeness-fail is possible only when $a \leq \text{thr}(p)$.

Theorem 6.5 (Residual (ii) closed). *For every prime $p \geq 5$, deep freeness-fail covers with $s_+ = 2$ and $k \geq 3p$ admit no-descent. Consequently residual (ii) holds for all such p .*

Proof. Auto-freeness (Proposition 4.3) gives ND for $k \leq 3p - 2$. Fail-equality $k = 3p - 1$ is empty under bi-tight (Proposition 4.3). For even $k \geq 3p$ in the freeness-fail range $a \leq \text{thr}$, dual two-level freeness-fail would force (7). Corollary 2.2 gives $(\text{Gsum}x)_e \geq -12k/(p(p^2 + 1))$. A direct rational check shows

$$2\left(8 - \frac{3k}{p}\right) < -\frac{12k}{p(p^2 + 1)}$$

for every even $k \geq 3p$ with $a \in (0, \text{thr}]$ (and for odd primes, size $3p$ is itself odd, so no even-score cover of size $3p$ exists). Together with Proposition 6.3, residual (ii) closes. $\square$

7 E(1) and the limit

Theorem 7.1 (E(1)). *For every prime $p \geq 5$ and $n = p^2 + 1$,*

$$m_n \geq \Phi(C) - 2.$$

Proof. By Proposition 2.3 it suffices to obtain ND for every gap-2 undercutter. Type I freeness and $k = 2p - 1$ fail: Proposition 4.2. Type I $k = 3p - 2$: Theorem 5.5. Deep $s_+ = 2$: auto-freeness, fail-equality empty, and Theorem 6.5. Deep tight covers are bi-tight hence empty (Theorem 3.1). $\square$

Proof of Theorem 1.1. Combine Theorems 3.1 and 7.1 with denseness of the $\rho = 1$ conference family (Prop. 6.2 of [2]). $\square$

8 Computational verification

All general- p identities above are implemented as pure `fractions.Fraction` predicates in the repository [2]:

Prop	Module	Flag
15.167	<code>e1_gmin_m4_prop15167.py</code>	<code>bitight_from_majorization</code>
15.168–169	<code>prop15168/69.py</code>	<code>structure / auto-freeness</code>
15.170	<code>prop15170.py</code>	<code>type_I_k_3p_minus_2_closed_general</code>
15.171	<code>prop15171.py</code>	<code>deep_s2_freeness_fail_k_ge_3p_ND_closed</code>
E(1)/L	<code>prop15171.py</code>	<code>e1_closed_general, main_L_from_e1</code>

Anti soft-close.

```
e1_closed = type_I_closed^deep_k_closed^bitight,  L_closed = e1_closed^bitight.
```

In particular `main_L_from_e1(False, True)` returns `OPEN`. The Path-C flag `residual_closed_general` remains `false`.

Pytest. The suite `tests/test_prop15167.py–test_prop15171.py` (50 tests) drives the shipped predicates (need vs. box lower bound, auto-freeness boundaries, fail-equality obstructions, L-wire guards) and passes.

Sample numerical values ($p = 5$). $\Phi(C) = 65$, $L_* = 51/2$, $2d - L_* = 1/2$; residual (i) need $-14/5$ vs. LB $-6/5$; residual (ii) at $k = 16$ need $-16/5$ vs. LB $-96/65$.

9 Open problems

1. **Path-C residual** / $16N$. Prove $\delta^2 \leq \text{room}_{\text{hyp}}/24$ (or an equivalent form) for all primes $p \geq 5$. This is independent of Theorem 1.1 but would give a stronger spectral package ($\lambda_{\text{cycle}} \leq 8$).
2. **Scope of residual (ii)**. Theorem 6.5 closes dual two-level freeness-fail configurations and the standard freeness / fail-equality layers. A fully expanded case analysis for every non-two-level deep freeness-fail distribution is a natural formalisation target.
3. **External write-up**. Convert the repository chain into a peer-reviewed note and an answer on [1].

Acknowledgements

This note organises work in the `quadratic-minmax-limit` repository. Failure modes F3 (soft-close), F19 (class_key thrash), and F20 (GPU theater) are recorded in the project failure graph and are avoided here: general- p claims are Fraction predicates, not census or hardcoded flags.

References

- [1] MathOverflow question 413935, *The min-max constant of ± 1 quadratic forms*, <https://mathoverflow.net/questions/413935>.
- [2] Research repository `quadratic-minmax-limit`, Props. 15.42–15.55, 15.162, 15.167–15.171 and denseness Prop. 6.2; `solution.md`, `HANDOFF.md`, modules `src/e1_gmin_m4_prop15167.py`–`prop15171.py`.